Notes: Beyhaghi, Fracassi, & Weitzner (JF, 2026)

Adverse Selection in Corporate Loan Markets

Contents

1 Big picture, research question, and mechanism

1.1 Why "more banks" can raise loan rates

In product markets, more sellers typically intensify competition and reduce prices. In loan markets, however, two informational frictions can reverse the comparative statics:

1. **Borrower adverse selection (borrower knows more than lenders).** Riskier borrowers have stronger incentives to seek funding and accept high rates.
2. **Lender-to-lender information asymmetry (some banks know more).** When a borrower approaches a bank, the bank may worry the borrower has already been rejected elsewhere. With more banks, this "winner's curse" problem can intensify (Broecker-style logic), producing (i) higher equilibrium rates, (ii) riskier funded borrowers conditional on observables, and (iii) higher aggregate lending volume.

1.2 Why adverse selection can generate *market power*

If some banks become *informed* about particular borrowers (screening advantages or relationship-based private information), then borrowers face a problem when trying to switch: an outside bank treats them as part of a pool and prices as if they might be "lemons". This generates **information rents** for the informed bank and implies markups can *increase* with the number of banks if adverse selection worsens.

2 Data and measurement

2.1 Supervisory micro data

The paper uses Federal Reserve Y-14Q Schedule H.1 supervisory filings, covering corporate loans above $1m made by large U.S. bank holding companies. The analysis focuses on **private, nonsyndicated borrowers**, where local information problems are plausibly strongest.

2.2 Key variables

- **Price:** loan interest rate, IR_l (in percent).
- **Banks' private risk assessments:**
  - probability of default PD_l (in percent),
  - loss given default LGD_l (in percent),
  - expected loss $\mathrm{EL}_l \equiv \mathrm{PD}_l \times \mathrm{LGD}_l$ (in percent).
- **Ex post performance:** nonperformance and realized default within 12 months.
- **Market structure:** number of banks in the county, NOB_c, defined as banks that extend a loan in that county at any point in the sample.
- **County controls:** population density, wages, finance-industry wages, population.

- **GSIB exposure:** number of GSIBs in county c in 2015, $\mathrm{NOG}_{2015,c}$, used for the surcharge shock.

2.3 How to read Table I and Table II

Table I (correlations). Different concentration measures are strongly related. In particular, loan HHI is tightly (negatively) correlated with the number of banks. This supports using NOB_c as a transparent "how many banks can a borrower approach" statistic.

Table I

Correlation between Measures of Market Concentration

This table contains a correlation matrix containing different measures of market concentration as well as the number of GSIBs present in 2015 at the county level.

Variables	(i)	(ii)	(iii)	(iv)	(v)
(i) Number of Banks	1.00				
(ii) Deposit HHI	−0.27	1.00			
(iii) Loan HHI	−0.90	0.23	1.00		
(iv) Number of GSIBs (2015)	0.79	−0.16	−0.77	1.00	
(v) Number of Banks (Branches)	0.84	−0.32	−0.69	0.70	1.00

Table II

Summary Statistics

This table contains summary statistics for loan level, firm, and geographic characteristics. The Appendix includes detailed variable definitions and Section II explains our filters.

	Mean	*SD*	10%	Median	90%	*N*
Loan Characteristics						
Amount (million USD)	7.17	14.74	1.06	2.66	16.19	21,924
Interest Rate (%)	3.66	1.17	2.16	3.66	5.24	21,924
Probability of Default (%)	1.34	1.70	0.20	0.87	2.79	21,924
Loss Given Default (%)	35.11	14.68	15.00	36.00	50.00	21,924
Expected Loss (%)	0.45	0.58	0.06	0.27	0.97	21,924
Floating Interest Rate	0.79	0.41	0.00	1.00	1.00	21,924
Guaranteed	0.50	0.50	0.00	0.00	1.00	21,924
Maturity (months)	41.70	30.25	11.00	36.00	84.00	21,924
Nonperformance (%)	2.05	14.18	0.00	0.00	0.00	21,924
Number of Prior Lenders	0.82	1.76	0.00	0.00	2.00	21,924
New Borrower	0.27	0.44	0.00	0.00	1.00	21,924
Line of Credit	0.50	0.50	0.00	0.00	1.00	21,924
Realized Default (%)	0.82	9.02	0.00	0.00	0.00	21,924
Secured	0.91	0.29	1.00	1.00	1.00	21,924
Secured by Blanket Lien	0.38	0.49	0.00	0.00	1.00	21,923
Stay Bank	0.76	0.43	0.00	1.00	1.00	16,087
GSIB	0.42	0.49	0.00	0.00	1.00	21,924
Firm Characteristics						
Assets (million USD)	153.50	706.61	4.30	23.59	207.67	21,924
Net Sales (million USD)	229.50	1,247.45	9.22	45.98	337.85	21,854
Leverage	0.34	0.25	0.02	0.31	0.68	21,480
Profitability	0.12	0.18	−0.00	0.07	0.28	21,924
Tangibility	0.91	0.17	0.67	0.99	1.00	21,866
Geographic Characteristics						
Number of Banks	11.64	5.86	4.00	12.00	20.00	21,924
Number of Banks (Annual)	5.74	3.77	1.00	5.00	11.00	21,924
Number of GSIBs (2015)	2.19	1.25	0.00	2.00	4.00	20,420
Number of All Banks	31.06	24.27	9.00	24.00	71.00	21,904
Population Density	6.83	1.54	4.72	7.05	8.30	21,924
Wages	9.50	0.27	9.19	9.48	9.81	21,907
Financial Industry Wages	9.84	0.39	9.39	9.81	10.28	21,901
Population	13.28	1.36	11.42	13.48	14.82	21,924
Deposit HHI	0.20	0.08	0.11	0.18	0.29	21,924
Loan HHI	0.50	0.25	0.22	0.43	0.91	21,924

Table II (summary statistics). The sample is dominated by relatively small private firms and modest-size loans, a setting in which local information frictions plausibly matter. The median loan rate is about 3.66% and the median PD is below 1%, with low realized default over one year.

3 Step 1: validating risk assessments as bank private information

The empirical strategy relies on PD and LGD being informative about risk *beyond* what the econometrician observes. The paper validates this by showing: (i) PD, LGD strongly predict interest rates, and (ii) once one controls for PD, LGD, interest rates do not predict subsequent default.

3.1 Model (1): pricing vs internal risk

The baseline pricing regression is:

$$\mathrm{IR}_l = \beta_0 \mathrm{PD}_l + \beta_1 \mathrm{LGD}_l + \beta_2 (\mathrm{PD}_l \times \mathrm{LGD}_l) + \Gamma X_l + \delta_{b,t} + \alpha_{i,t} + u_l, \tag{1}$$

where l indexes loans, b banks, t quarters, and i industries. X_l includes loan controls (maturity, amount, guarantee, loan type/purpose), $\delta_{b,t}$ are bank×quarter fixed effects (absorbing bank funding costs and risk-model differences), and $\alpha_{i,t}$ are industry×quarter fixed effects. Standard errors are clustered by county.

Table III (interpretation). PD, LGD, and expected loss significantly predict interest rates. The adjusted R^2 rises when risk assessments are included, and the PD coefficient implies economically meaningful sensitivity of pricing to perceived risk.

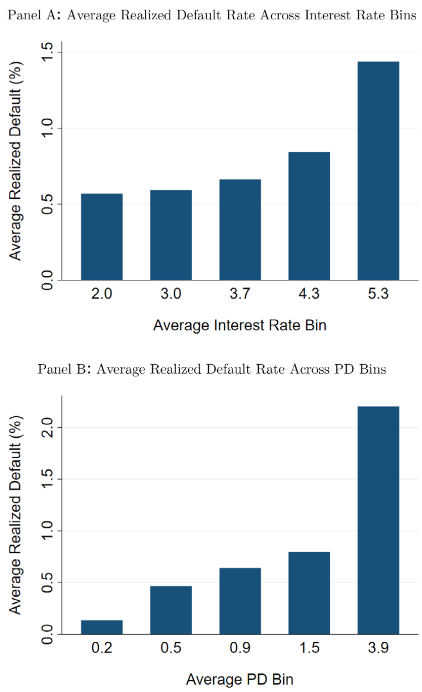

Figure 1. Average realized default rates. Panel A plots the average realized default rates over the 12 months following origination across five interest rate bins. Panel B plots the average realized default rates across five PD bins. The average interest rate or PD in each bin is listed below each bar. (Color figure can be viewed at wileyonlinelibrary.com)

Table III
Interest Rates and Risk Assessments

This table tests whether banks' internal risk assessments predict loan interest rates. t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Interest Rate (%)	
	(1)	(2)
Probability of Default (%)		0.077***
		(4.660)
Loss Given Default (%)		0.003***
		(4.366)
Expected Loss (%)		0.155***
		(3.542)
Log(maturity)	−0.001	0.016*
	(0.154)	(1.842)
Log(Amount)	−0.159***	−0.151***
	(20.598)	(20.563)
Guaranteed	0.062***	0.061***
	(3.255)	(3.416)
Loan Purpose FE	YES	YES
Loan Type FE	YES	YES
Bank-Quarter FE	YES	YES
Industry-Quarter FE	YES	YES
Observations	21,853	21,853
Adj. R^2	0.52	0.55

3.2 Figure 1: PD bins predict default more cleanly than rate bins

Figure 1 compares realized default rates across (i) interest-rate bins and (ii) PD bins. The PD-bin relationship is monotone and steep, while rate bins show a weaker and less monotone association. **Interpretation:** interest rates embed nonrisk components (markups), whereas PD captures risk more directly.

3.3 Model (2): loan performance vs rate and risk

To test whether rates contain residual risk information beyond bank assessments, the paper estimates:

$$y_l = \beta_0 \mathrm{IR}_l + \beta_1 \mathrm{PD}_l + \beta_2 \mathrm{LGD}_l + \beta_3(\mathrm{PD}_l \times \mathrm{LGD}_l) + \Gamma X_l + \delta_{b,t} + \alpha_{i,t} + u_l, \tag{2}$$

where y_l is either nonperformance or realized default within one year.

Table IV (interpretation). Interest rates predict future nonperformance/default *when alone*. But once PD, LGD are included, the interest-rate coefficient becomes small and statistically insignificant, while PD remains a strong predictor. **Key inference:** conditional on PD, LGD, rate variation is plausibly "markup-like" rather than residual risk.

4 Step 2: market structure, adverse selection, and quantities

4.1 Model (3): interest rates vs number of banks

The main market-structure regression is:

$$\mathrm{IR}_l = \beta \mathrm{NOB}_c + \Gamma_0 X_l + \Gamma_1 Z_{f,t} + \delta_{b,t} + \alpha_{i,t} + u_l, \tag{3}$$

where c is the borrower county and $Z_{f,t}$ are firm controls (size, leverage, tangibility, profitability).

Table IV

Interest Rates, Risk Assessments, and Loan Performance

This table tests whether interest rates and banks' internal risk assessments predict nonperformance and default after controlling for loan characteristics and fixed effects. t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Nonperformance (%)			Realized Default (%)		
	(1)	(2)	(3)	(4)	(5)	(6)
Interest Rate (%)	0.527***		0.101	0.354***		0.093
	(3.699)		(0.541)	(3.346)		(0.620)
Probability of Default (%)		1.392**	1.384**		0.888***	0.880**
		(2.259)	(2.206)		(2.624)	(2.524)
Loss Given Default (%)		0.028	0.028		0.014	0.014
		(1.501)	(1.450)		(1.469)	(1.375)
Expected Loss (%)		−1.229	−1.244		−0.816	−0.830
		(0.918)	(0.940)		(1.262)	(1.307)
Loan Controls	YES	YES	YES	YES	YES	YES
Bank-Quarter FE	YES	YES	YES	YES	YES	YES
Industry-Quarter FE	YES	YES	YES	YES	YES	YES
Observations	18,246	18,246	18,246	18,246	18,246	18,246
Adj. R^2	0.09	0.10	0.10	0.07	0.08	0.08

Table V

Market Structure and Interest Rates

This table tests the relationship between the number of banks and interest rates. t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Interest Rate (%)		
	(1)	(2)	(3)
Number of Banks	0.012***	0.013***	0.009**
	(6.338)	(6.318)	(2.433)
Log(Assets)		−0.151***	−0.152***
		(20.304)	(20.735)
Leverage		0.204***	0.207***
		(6.366)	(6.559)
Tangibility		−0.671***	−0.672***
		(14.876)	(14.922)
Profitability		−0.388***	−0.382***
		(9.027)	(9.070)
Population Density			−0.008
			(0.369)
Wages			0.142
			(1.479)
Financial Industry Wages			0.037
			(0.569)
Loan Controls	YES	YES	YES
Bank-Quarter FE	YES	YES	YES
Industry-Quarter FE	YES	YES	YES
Observations	21,853	21,388	21,348
Adj. R^2	0.52	0.54	0.54

Table V (interpretation). The coefficient on NOB_c is positive and statistically significant across specifications. Even after adding county-level cost proxies (population density and wage measures), counties with more banks have higher loan rates.

Figure 2 (Nonmonotonicity). See Figure 2 at the end of the document for a cleaner layout. Using bank-count groups, Figure 2 shows a **U-shaped** pattern: rates fall from one bank to a few banks (competition effect) and then rise in high-bank markets (adverse-selection effect).

4.2 Borrower risk: PD vs number of banks (Table VI and Figure 3)

Replacing IR_l with PD_l as the dependent variable yields a positive relationship: borrowers funded in high-bank markets have higher internal PDs, conditional on observables. Figure 3 shows PD increasing monotonically with bank-count groups. **Interpretation:** more banks increases the probability that riskier borrowers obtain funding.

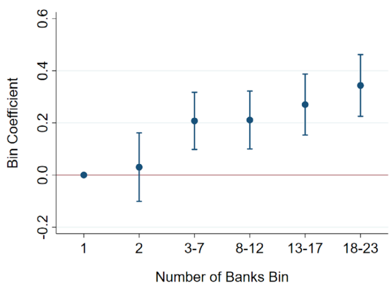

Figure 3. Number of banks in the county and PDs. This figure plots estimated coefficients, with 90% confidence intervals, for regressions of PD on different number of bank group dummies as well as the same controls as in column (2) of Table VI. The number of banks in each group is listed below, where the reference number of banks is one. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

Table VI

Market Structure and Borrower Risk

This table tests the relationship between the number of banks and probability of default (PD). t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Probability of Default (%)		
	(1)	(2)	(3)
Number of Banks	0.008***	0.011***	0.009**
	(3.470)	(4.970)	(2.330)
Log(Assets)		−0.138***	−0.137***
		(10.845)	(10.764)
Leverage		0.963***	0.961***
		(11.562)	(11.648)
Tangibility		−0.235***	−0.232***
		(2.757)	(2.715)
Profitability		−1.822***	−1.828***
		(24.588)	(24.871)
Population Density			0.045***
			(3.164)
Wages			−0.169
			(1.536)
Financial Industry Wages			−0.034
			(0.327)
Loan Controls	YES	YES	YES
Bank-Quarter FE	YES	YES	YES
Industry-Quarter FE	YES	YES	YES
Observations	21,853	21,388	21,348
Adj. R^2	0.17	0.23	0.23

4.3 Model (4): quantities at the county level

A key prediction is that aggregate lending volume rises with the number of banks when adverse selection is present. The paper aggregates to county-quarter and estimates:

$$\text{Volume}_{c,t} = \beta_0 \text{NOB}_c + \Gamma X_{c,t} + \delta_t + u_{c,t}, \tag{4}$$

where $\text{Volume}_{c,t}$ is log loan volume (or number of loans), $X_{c,t}$ are county controls, and δ_t are quarter fixed effects.

Table VII and Figure 4 (interpretation). Loan volume is increasing in NOB_c and the grouped estimates are monotone. **Joint reading with rates and PDs:** high-bank markets have *more lending* but also *higher rates* and *riskier funded borrowers*, a pattern consistent with adverse selection and inconsistent with standard competition alone.

Table VII

Market Structure and Loan Volume

This table tests the relationship between the number of banks and loan volume. Loan volume is measured in logs and is aggregated at the county-quarter level. t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Log(Loan Volume)		
	(1)	(2)	(3)
Number of Banks	0.144***	0.140***	0.125***
	(21.235)	(14.662)	(11.707)
Population Density		−0.040*	−0.065**
		(1.821)	(2.535)
Wages		0.161	0.213
		(0.945)	(1.255)
Financial Industry Wages		0.173*	0.160
		(1.766)	(1.624)
Population			0.086**
			(2.164)
Quarter FE	YES	YES	YES
Observations	8,060	8,026	8,026
Adj. R^2	0.26	0.27	0.27

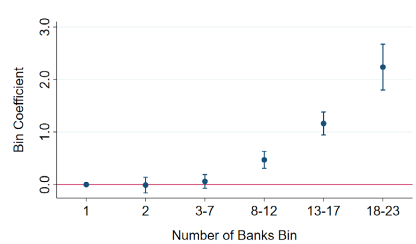

Figure 4. Number of banks in the county and loan volume. This figure plots estimated coefficients, with 90% confidence intervals, for regressions of loan volume on different number of bank group dummies as well as the same controls as in column (2) of Table VII. The number of banks in each group is listed below, where the reference number of banks is one. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

4.4 Ancillary evidence: PD accuracy, collateralization, and switching

Figure 5 (ROC/AUC). PDs are more predictive of realized outcomes in markets with more banks (higher AUC). **Interpretation:** when the winner's curse is stronger, "winning" a borrower reveals more about its type, improving the informativeness of internal PD.

Table VIII (blanket liens). Loans are more likely to be secured by blanket liens in markets with more banks. **Interpretation:** collateral is used to mitigate screening/information problems as adverse selection worsens.

Table VIII

Market Structure and Collateralization

This table tests the relationship between the number of banks and whether loans contain a blanket lien. *t*-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Secured by Blanket Lien		
	(1)	(2)	(3)
Number of Banks	0.002***	0.002***	0.002**
	(2.977)	(3.116)	(2.246)
Log(Assets)		−0.017***	−0.017***
		(7.530)	(7.535)
Leverage		−0.033***	−0.034***
		(2.685)	(2.743)
Tangibility		−0.076***	−0.078***
		(4.070)	(4.192)
Profitability		0.078***	0.076***
		(4.999)	(4.904)
Population Density			−0.003
			(0.656)
Wages			−0.029
			(0.976)
Financial Industry Wages			0.023
			(1.143)
Loan Controls	YES	YES	YES
Bank-Quarter FE	YES	YES	YES
Industry-Quarter FE	YES	YES	YES
Observations	21,853	21,388	21,348
Adj. R^2	0.56	0.56	0.56

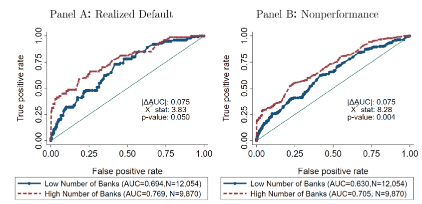

Figure 5. Market structure and PD accuracy. Panel A plots ROC curves split by whether the number of banks is above the median over the sample, with PD the predictor and realized default the outcome variable. Panel B uses PD as the predictor and nonperformance as the outcome variable. The area under each ROC curve (AUC) is reported along with the number of observations in the legend. |ΔAUC| reports the difference between the two AUCs. Below |ΔAUC|, DeLong, DeLong, and Clarke-Pearson (1988) statistics are reported: the χ^2-test statistic and its corresponding *p*-value are reported, which test the null hypothesis that the difference between the two AUC values equals zero. (Color figure can be viewed at wileyonlinelibrary.com)

Figures 6 and 7 (staying with the same bank). Firms stay with existing banks about three-quarters of the time, with only minor changes as the number of banks increases. **Interpretation:** even in seemingly competitive markets, information-based frictions can impede switching, setting up informational market power.

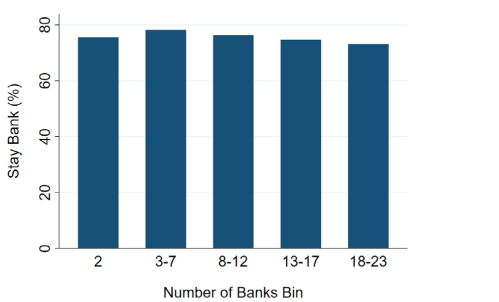

Figure 6. The frequency of firms staying with their existing banks. This figure plots the frequency of firms staying with one of their existing banks, that is, *Stay Bank* across counties with different numbers of banks. Loans from counties with one bank are excluded. (Color figure can be viewed at wileyonlinelibrary.com)

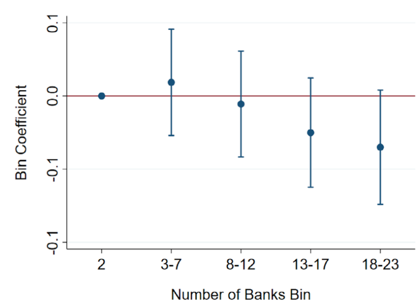

Figure 7. The frequency of firms staying with their existing banks (regression analysis). This figure plots estimated coefficients, with 90% confidence intervals, for regressions of *Stay Bank* on number of bank group dummies as well as the same controls as in column (2) of Table VI. The number of banks in each group is listed below, where the reference number of banks is two. Loans from counties with one bank are excluded. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

5 Step 3: markups and market power

5.1 Markup idea

Write a loan rate schematically as:

$$\mathrm{IR} = \underbrace{\text{funding/processing costs}}_{\text{absorbed by bank}\times\text{t and county controls}} + \underbrace{\text{risk component}}_{\text{captured by PD,LGD,EL}} + \underbrace{\text{markup}}_{\text{residual}}.$$

The paper operationalizes "markup" as interest-rate variation *after* controlling for the bank's perceived risk.

5.2 Model (5): risk-orthogonalized rate vs market structure

$$\mathrm{IR}_l = \beta_0 \mathrm{NOB}_c + \beta_1 \mathrm{PD}_l + \beta_2 \mathrm{LGD}_l + \beta_3(\mathrm{PD}_l \times \mathrm{LGD}_l) + \Gamma_0 X_l + \Gamma_1 Z_{f,t} + \delta_{b,t} + \alpha_{i,t} + u_l. \quad (5)$$

Table IX (interpretation). NOB_c remains positive and significant even after controlling for $\mathrm{PD}, \mathrm{LGD}, \mathrm{EL}$. **Interpretation:** markets with more banks exhibit *higher markups* (relative pricing) on observationally comparable loans, consistent with adverse selection generating informational market power.

Table IX

Market Structure and Markups

This table tests the relationship between the number of banks and markups, where we consider markups as any variation in interest rates after controlling for the risk of the loan. t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Interest Rate (%)		
	(1)	(2)	(3)
Number of Banks	0.011***	0.012***	0.008**
	(5.948)	(5.844)	(2.343)
Probability of Default (%)	0.076***	0.064***	0.064***
	(4.615)	(3.875)	(3.988)
Loss Given Default (%)	0.003***	0.002***	0.002***
	(4.360)	(2.962)	(2.950)
Expected Loss (%)	0.155***	0.143***	0.142***
	(3.532)	(3.303)	(3.330)
Log(Assets)		−0.132***	−0.133***
		(17.901)	(18.270)
Leverage		0.108***	0.111***
		(3.474)	(3.630)
Tangibility		−0.624***	−0.625***
		(14.289)	(14.322)
Profitability		−0.200***	−0.192***
		(4.721)	(4.627)
Population Density			−0.012
			(0.620)
Wages			0.165*
			(1.789)
Financial Industry Wages			0.032
			(0.494)
Loan Controls	YES	YES	YES
Bank-Quarter FE	YES	YES	YES
Industry-Quarter FE	YES	YES	YES
Observations	21,853	21,388	21,348
Adj. R^2	0.55	0.56	0.56

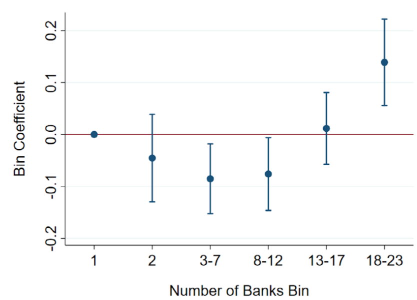

Figure 8. Number of banks in the county and markups. This figure plots estimated coefficients, with 90% confidence intervals, for regressions of interest rate on number of bank group dummies as well as the same controls as in column (2) of Table IX. We refer to markups as any variation in interest rates after controlling for the risk of the loan. The number of banks in each group is listed below, where the reference number of banks is one. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 8. The grouped markup effects are U-shaped, mirroring interest rates: initial competition lowers markups, but markups rise once adverse selection dominates in high-bank markets.

6 Cross-sectional mechanism check: firm tangibility (Table X)

If adverse selection is fundamentally about unobserved borrower quality, it should be stronger when opacity is higher. The paper splits the sample by tangibility each quarter.

Table X

Adverse Selection and Firm Tangibility

This table tests the relationship between the number of banks and interest rates, PDs and markups for high- and low-tangibility firms. In each quarter, we sort loan observations into two groups based on borrower tangibility: above-median (high-tangibility) and below-median (low-tangibility). t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Interest Rate (%)		Probability of Default (%)		Interest Rate (%)	
	High Tangibility	Low Tangibility	High Tangibility	Low Tangibility	High Tangibility	Low Tangibility
	(1)	(2)	(3)	(4)	(5)	(6)
Number of Banks	0.003	0.013***	0.000	0.014**	0.003	0.011***
	(0.681)	(3.769)	(0.101)	(2.466)	(0.808)	(3.234)
Probability of Default (%)					0.051***	0.101***
					(2.653)	(6.528)
Loss Given Default (%)					0.003***	0.003***
					(2.856)	(2.841)
Expected Loss (%)					0.140***	0.087*
					(2.700)	(1.869)
Log(Assets)	−0.203***	−0.105***	−0.161***	−0.127***	−0.184***	−0.086***
	(17.683)	(10.326)	(7.647)	(8.672)	(17.435)	(8.336)
Leverage	0.170***	0.279***	1.064***	0.703***	0.079*	0.193***
	(3.729)	(6.638)	(9.905)	(8.736)	(1.808)	(4.652)
Tangibility	7.593	−0.680***	23.017**	−0.345***	5.867	−0.622***
	(1.309)	(12.106)	(2.221)	(3.998)	(1.001)	(11.449)
Profitability	−0.257***	−0.624***	−1.594***	−2.251***	−0.114**	−0.348***
	(4.990)	(8.981)	(17.261)	(17.249)	(2.307)	(4.617)
Population Density	−0.000	−0.016	0.057**	0.020	−0.006	−0.018
	(0.022)	(0.843)	(2.558)	(1.066)	(0.297)	(0.992)
Wages	0.080	0.221**	−0.129	−0.124	0.100	0.235**
	(0.735)	(1.995)	(0.804)	(0.799)	(0.945)	(2.264)

Table X—*Continued*

	Interest Rate (%)		Probability of Default (%)		Interest Rate (%)	
	High Tangibility	Low Tangibility	High Tangibility	Low Tangibility	High Tangibility	Low Tangibility
	(1)	(2)	(3)	(4)	(5)	(6)
Financial Industry Wages	0.086	−0.038	−0.177	0.029	0.088	−0.041
	(1.234)	(0.458)	(1.023)	(0.268)	(1.251)	(0.546)
Loan Controls	YES	YES	YES	YES	YES	YES
Bank-Quarter FE	YES	YES	YES	YES	YES	YES
Industry-Quarter FE	YES	YES	YES	YES	YES	YES
Observations	10,563	10,709	10,563	10,709	10,563	10,709
Adj. R^2	0.55	0.56	0.32	0.19	0.57	0.59

Table X (interpretation). For **high-tangibility** firms, the relationships between number of banks and (i) rates, (ii) PD, and (iii) markups are near zero. For **low-tangibility** firms, the effects are positive and stronger than baseline. **Mechanism inference:** the market-structure patterns are driven by the part of the market where asymmetric information is most severe.

7 Relationship lending and information rents (Model (6) and Table XI)

7.1 Model (6): stay with existing bank and markups

Restrict to borrowers with repeat borrowing (loans after the first observed loan). Estimate:

$$\text{IR}_l = \beta_0 \text{StayBank}_l + \beta_1 \text{PD}_l + \beta_2 \text{LGD}_l + \beta_3 (\text{PD}_l \times \text{LGD}_l) + \Gamma_0 X_l + \Gamma_1 Z_{f,t} + \delta_{b,t} + \alpha_{i,t} + \lambda_{c,t} + u_l, \quad (6)$$

where $\lambda_{c,t}$ are county×quarter fixed effects. With $\text{PD}, \text{LGD}, \text{EL}$ included, β_0 is interpreted as a **markup difference** between staying vs switching.

Table XI (interpretation). Staying with the same bank is associated with higher markups (about 9 bps in the baseline controlled specification). Moreover, the interaction with the number of prior lenders is negative: the informational holdup is weaker when borrowers have more past lending relationships (more informed outside options). **Interpretation:** relationship-based private information can create market power and rents.

Table XI

Switching Banks and Markups

This table tests whether firms that stay with their existing banks face higher markups, where we consider markups as any variation in interest rates after controlling for the risk of the loan. t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Interest Rate (%)		
	(1)	(2)	(3)
Stay Bank	0.040	0.090***	0.160***
	(1.647)	(3.637)	(4.925)
N. of Prior Lenders			0.037***
			(2.701)
Stay Bank × N. of Prior Lenders			−0.052***
			(3.504)
Probability of Default (%)	0.102***	0.089***	0.088***
	(4.522)	(4.003)	(3.945)
Loss Given Default (%)	0.006***	0.005***	0.005***
	(5.047)	(4.654)	(4.630)
Expected Loss (%)	0.061	0.044	0.045
	(0.954)	(0.706)	(0.724)
Log(Assets)		−0.131***	−0.127***
		(12.135)	(10.789)
Leverage		0.187***	0.193***
		(4.110)	(4.246)
Tangibility		−0.677***	−0.676***
		(9.379)	(9.300)
Profitability		−0.288***	−0.285***
		(5.006)	(5.010)
Loan Controls	YES	YES	YES
County-Quarter FE	YES	YES	YES
Bank-Quarter FE	YES	YES	YES
Industry-Quarter FE	YES	YES	YES
Observations	12,257	11,915	11,915
Adj. R^2	0.61	0.62	0.62

Table XII

GSIB Surcharges and Market Outcomes (Reduced-Form Difference-in-Differences)

This table contains reduced-form difference-in-differences regressions testing whether the number of GSIBs induces changes in market outcomes following the imposition of capital surcharges. The sample period is 2014Q4 to 2019Q4 except for column (1) which is 2015Q1 to 2019Q4 in order to calculate the number of banks based on a full year. t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	N. of Banks	Loan Volume (log)	Interest Rate (%)	Probability of Default (%)	Interest Rate (%)
	(1)	(2)	(3)	(4)	(5)
Post × N. of GSIBs (2015)	−0.201**	−0.084***	−0.031*	−0.069*	−0.021
	(2.443)	(3.158)	(1.957)	(1.676)	(1.317)
Probability of Default (%)					0.078***
					(3.756)
Loss Given Default (%)					0.005***
					(4.592)
Expected Loss (%)					0.103*
					(1.771)
Loan Controls	YES	YES	YES	YES	YES
Bank-County FE	YES	YES	YES	YES	YES
Bank-Quarter FE	YES	YES	YES	YES	YES
Industry-Quarter FE	YES	YES	YES	YES	YES
Observations	12,697	13,640	13,640	13,640	13,640
Adj. R^2	0.91	0.72	0.61	0.29	0.63

8 Quasi-exogenous variation: GSIB capital surcharges (Models (7), Table XII, Figures 9–14, Table XIII)

8.1 Institutional shock

Capital surcharges on GSIBs were approved in 2015 and implemented starting in 2016 (phased in through 2019), raising lending costs for GSIBs and plausibly reducing their local lending presence.

sectionModel (7): Bartik-style difference-in-differences

$$y_{t,c} = \beta_0 + \beta_1 \Big(\text{NOG}_{2015,c} \times \text{Post}_t\Big) + \Gamma X + \delta_{b,t} + \gamma_{b,c} + \alpha_{i,t} + u_{t,c}, \tag{7}$$

where $\text{Post}_t = \mathbf{1}\{t \geq 2016\}$ and $y_{t,c}$ is either the (annual) number of banks, loan volume, interest rates, PDs, or markups.

Table XII (reduced-form interpretation). Counties with higher GSIB presence in 2015 experience (i) fewer banks and lower lending volume after 2016, and (ii) lower interest rates and lower PDs (and somewhat lower markups). **Interpretation:** a forced reduction in high-bank environments reduces adverse selection pressure and improves the average funded borrower pool.

Figures 9–14 (dynamic coefficients). The event-study style plots show post-2016 declines in bank presence, lending volume, rates, PDs, and markups in high-GSIB counties, with no visually obvious pre-trends for outcomes where pre-period data exist.

8.2 2SLS logic (Table XIII)

The paper then instruments the annual number of banks with $\text{NOG}_{2015,c} \times \text{Post}_t$ and estimates effects on: loan volume, interest rates, PDs, and markups.

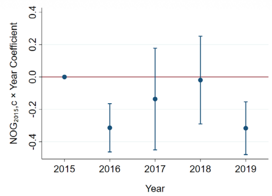

Figure 9. The effect of GSIB surcharges on the number of banks. This figure plots estimated regression coefficients with 90% confidence intervals from a version of (7) with annual interaction terms and the number of banks (calculated annually) as the dependent variable. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

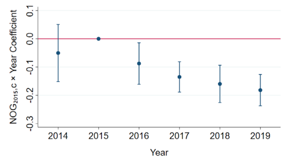

Figure 10. The effect of GSIB surcharges on lending volume. This figure plots estimated regression coefficients with 90% confidence intervals from a version of (7) with annual interaction terms and the log of loan volume as the dependent variable. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

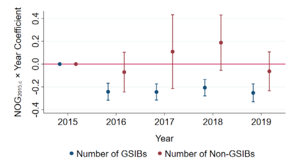

Figure 11. The effect of GSIB surcharges on the number of GSIB and non-GSIBs. This figure plots estimated regression coefficients with 90% confidence intervals from a version of (7) with annual interaction terms and either the number of GSIBs or non-GSIBs (both calculated annually) as the dependent variable. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

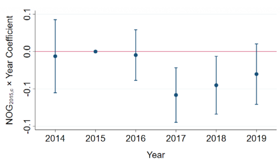

Figure 12. The effect of GSIB surcharges on interest rates. This figure plots estimated regression coefficients with 90% confidence intervals from a version of (7) with annual interaction terms and interest rate (%) as the dependent variable. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

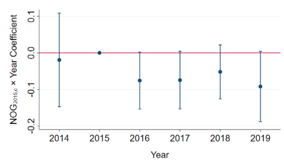

Figure 13. The effect of GSIB surcharges on PDs. This figure plots estimated regression coefficients with 90% confidence intervals from a version of (7) with annual interaction terms and probability of default (%) as the dependent variable. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

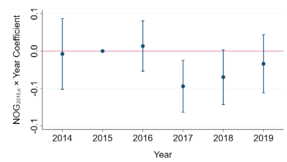

Figure 14. The effect of GSIB surcharges on markups. This figure plots estimated regression coefficients with 90% confidence intervals from a version of (7) with annual interaction terms and interest rate (%) as the dependent variable, while controlling for the risk of the loan (i.e., PD, LGD, and expected loss). Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

Table XIII (2SLS interpretation). Instrumented increases in the number of banks raise (i) loan volume, (ii) interest rates, and (iii) borrower risk (PD), while the markup estimate is positive but not statistically precise. IV estimates are larger than OLS, consistent with (a) measurement noise in bank counts and (b) the shock shifting bank presence mainly in high-bank counties where adverse selection dominates.

Table XIII

GSIB Surcharges and Market Outcomes (Two-Stage Least Squares)

This table contains two-stage least-squares regressions of market outcomes on the number of banks in the county. The excluded instrument is Number of GSIBs $(2015)_c$ × Post$_c$. The first stage is shown in column (1) of Table XII. The sample period is 2015Q1 to 2019Q4 in all specifications. t-Statistics are reported below the parameter estimates in parentheses and are calculated using robust standard errors clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Loan Volume (log)	Interest Rate (%)	Probability of Default (%)	Interest Rate (%)
(1)	(2)	(3)	(4)	(5)
Number of Banks (Annual)	0.525**	0.179*	0.396*	0.119
	(2.345)	(1.851)	(1.664)	(1.373)
Probability of Default (%)				0.082***
				(3.911)
Loss Given Default (%)				0.004***
				(4.442)
Expected Loss (%)				0.091
				(1.583)
Loan Controls	YES	YES	YES	YES
Bank-County FE	YES	YES	YES	YES
Bank-Quarter FE	YES	YES	YES	YES
Industry-Quarter FE	YES	YES	YES	YES
Observations	12,697	12,697	12,697	12,697

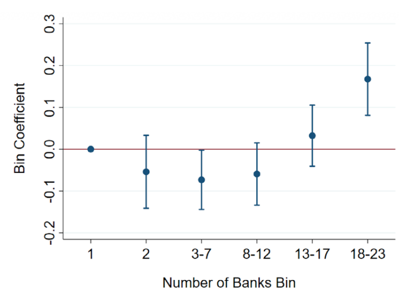

Figure 2. Number of banks in the county and interest rates. This figure plots estimated coefficients, with 90% confidence intervals, for regressions of interest rates on different number of bank group dummies as well as the same controls as in column (2) of Table V. The number of banks in each group is listed below, where the reference number of banks is one. Standard errors are clustered by county. (Color figure can be viewed at wileyonlinelibrary.com)

9 Conclusion: what to conclude

1. **Measurement contribution:** bank-internal PD and LGD provide an unusually direct proxy for lenders' private risk information.

2. **Core empirical pattern:** more banks ⇒ higher rates, higher borrower risk, and higher lending volume (conditional on observables).

3. **Market power result:** after controlling for private risk assessments, markups are higher in high-bank markets and higher for relationship (stay-bank) loans.

4. **Causal support:** GSIB surcharge shock reduces bank presence in GSIB-heavy counties and is associated with lower rates and lower borrower risk.

10 Conclusion

10.1 Summary

This paper concludes that **local corporate loan markets can exhibit "reversed" competition effects when adverse selection is severe**. In particular, markets with **more banks** can feature **higher loan interest rates**, **riskier funded borrowers**, and **greater lending volume**—patterns that are difficult to reconcile with standard models in which more competition only compresses margins.

The paper's contribution is a mechanism-and-measurement chain:

(i) **Measurement contribution: banks' internal risk assessments reveal private information.** Using supervisory data, the authors show that banks' internal PD and LGD strongly predict pricing, and that once PD and LGD are controlled for, interest rates no longer predict subsequent default/nonperformance. This supports interpreting residual rate variation as *markup-like* rather than unobserved risk.

(ii) **Core cross-sectional fact: more banks $\Rightarrow$ higher rates, higher risk, higher volume.** Conditional on loan and firm characteristics and rich fixed effects, the interest rate rises with the number of banks in a county. Borrowers that receive loans in high-bank counties have higher internal PD. Loan volume (at the county level) is also increasing in the number of banks.

(iii) **Market power result: adverse selection can *create* markups.** After controlling for banks' private risk assessments, **markups are higher in high-bank markets**. Moreover, firms that **stay with their existing bank** face higher markups (information rents / hold-up), and the hold-up attenuates when firms have more prior lenders (more informed outside options).

(iv) **Causal support: a plausibly exogenous shock to bank presence moves outcomes in the adverse-selection direction.** The GSIB capital surcharges (implemented from 2016 onward) reduce bank presence and lending in GSIB-exposed counties and are associated with **lower interest rates and lower borrower risk**, supporting the interpretation that adverse selection can dominate standard competition effects.

10.2 Key empirical conclusions

From the main tables are:

- **Rates increase with the number of banks (Table V).** The coefficient is about 0.012 (in percentage points), i.e. **1.2 bps per additional bank**, and remains positive with firm controls and county cost proxies.

- **Borrower risk increases with the number of banks (Table VI).** The coefficient on the number of banks in the PD regression is positive (e.g. about 0.011 percentage points in a benchmark specification).

- **Quantity increases with the number of banks (Table VII).** County-quarter log loan volume rises with the number of banks (e.g. coefficient ≈ 0.144 in a benchmark specification).

- **Risk-orthogonal markups increase with the number of banks (Table IX).** After controlling for PD and LGD, the number-of-banks coefficient remains about 0.012 percentage points (again **roughly 1.2 bps per additional bank**), consistent with informational market power.

- **Relationship lending: staying with the same bank is associated with higher markups (Table XI).** In controlled specifications with county$\times$quarter fixed effects, **Stay Bank** is about **9 bps** higher markup. The interaction **Stay Bank**$\times$(# prior lenders) is negative, consistent with weaker hold-up when more banks are already informed.

- **GSIB surcharge shock (Table XII).** Counties with more GSIB presence pre-2016 experience declines in bank presence and loan volume post-2016, and also see **lower interest rates and lower borrower risk** (a pattern aligned with the adverse-selection channel).

- **2SLS magnitudes (Table XIII).** Instrumenting bank presence with GSIB exposure, one additional annual bank raises: **interest rates by about 18 bps** and PD **by about 40 bps**; volume rises substantially as well, while the markup estimate is positive but less precisely estimated.

10.3 Why *more banks* can raise *rates* and *markups*

It combines a **winner's-curse adverse selection effect** with **informational hold-up**.

1) Winner's-curse adverse selection (cross-sectional equilibrium). When borrowers can approach many banks, banks rationally infer that a borrower who shows up at their door may have already been screened by others. If other banks reject bad borrowers, then the pool of borrowers who keep shopping is *negatively selected*. As the number of banks increases, this adverse selection can intensify:

$$\text{more banks} \;\Rightarrow\; \text{more screening and more rejection elsewhere} \;\Rightarrow\; \text{worse pool conditional on applying.}$$

To break even on a worse pool, banks raise equilibrium rates and require stronger protection (e.g. collateral), and the funded set becomes riskier.

2) Information rents and switching frictions (markups). Relationship lenders can have private information about a borrower (via monitoring and repeated interaction). When the borrower tries to switch, outside banks treat the borrower as part of a pool and price defensively (a "lemons" discount), which lowers the borrower's outside option. This gives the incumbent bank market power *even in a market with many banks*:

$$\text{outside banks are uninformed} \Rightarrow \text{switching is costly} \Rightarrow \text{incumbent extracts a markup.}$$

The paper's evidence that **Stay Bank** predicts higher risk-adjusted rates, and that the hold-up weakens with more prior lenders, matches this informational-rent channel.

3) Why the relationship can be nonmonotone (U-shape). With few banks, standard competition effects dominate: adding a bank increases competitive pressure and can reduce rates. With many banks, the marginal impact on competition diminishes, while adverse selection and informational frictions can rise, so rates and markups can increase—producing the empirically observed U-shape in grouped plots.

10.4 Interpretation and scope (what the paper does *not* claim)

The paper does not present a full welfare analysis of market structure. Its central message is positive (not normative): **market structure interacts with informational frictions**, so the sign of "competition" effects in credit markets cannot be inferred from standard product-market logic alone. The evidence implies a trade-off: more banks may expand credit access/volume but can also raise rates and risk through adverse selection.

10.5 Takeaway

Using supervisory loan data with banks' internal risk assessments, the paper shows that in corporate loan markets adverse selection can dominate standard competition: counties with more banks have higher interest rates, higher borrower risk, higher loan volume, and higher risk-adjusted markups (especially for relationship loans), and a GSIB-surcharge shock that reduces bank presence lowers rates and borrower risk—highlighting that information frictions can generate market power and reverse standard market-structure predictions.